

Skew contractions and exact weighted Pauli uncertainty on all line graphs*

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Abstract

For a family of Pauli observables, let its frustration graph join exactly the anticommuting pairs. We prove that the complete weighted quadratic uncertainty problem is exact on every line graph. More precisely, for every finite simple graph R and every nonnegative edge weight w ,

$$\beta(L(R), w) = \alpha(L(R), w) = \nu(R, w),$$

where $\nu(R, w)$ is the maximum weight of a matching of R . The main proof maps the Hamiltonian to a Majorana bilinear. Nuclear-norm duality produces a skew contraction Q , and the squared supported entries Q_e^2 satisfy both the degree and all odd-set inequalities of Edmonds' matching polytope. It also yields the sharp matrix inequality

$$(\|A\|_*/2)^2 \leq \nu(R, w) \sum_{e \in E(R)} A_e^2/w_e$$

for every real skew matrix supported on R . A second proof derives the same matching-polytope constraints from simplicial claw-free rank inequalities. The family $L(K_{2k+1})$, $k \geq 2$, proves that the result strictly extends the previously sufficient h -perfect route. A centralized ablation over all 1245 nonempty graph-atlas roots through seven vertices and 6225 weighted instances separates the ingredients: degree constraints are exact on 66.89%, cliques plus odd cycles on 84.69%, and the complete odd-set system on 100%. All three are exact on 710 bipartite controls. A symmetric norm-one contraction violates a triangle odd-set inequality, while its skew control saturates it. These computations test the mechanism; the theorem remains analytic.

Status and scope. This is a self-contained working manuscript, not an externally reviewed article. The proof uses classical matching-polytope and Majorana tools, which are cited explicitly. A documented search found no prior statement of the all-weight equality above, but a negative literature search is not a proof of priority. No quantum advantage or new matching algorithm is claimed.

1 Introduction

Let $P_1, \dots, P_n$ be Hermitian Pauli observables. Their frustration graph G has vertex set $[n]$ and an edge ij exactly when $P_i P_j = -P_j P_i$; nonedges commute. For nonnegative w , define

$$\beta(G, w) = \sup_{\{P_i\}, \rho} \sum_i w_i \operatorname{tr}(\rho P_i)^2, \quad \alpha(G, w) = \max_{S \text{ stable in } G} \sum_{i \in S} w_i. \quad (1)$$

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The supremum uses Pauli realizations with exactly the prescribed pairwise relations. Representation invariance of this graph parameter is established in the beta-number framework of Xu et al. [4]. Since the observables in a stable set commute, a common eigenstate gives $\beta(G, w) \geq \alpha(G, w)$. The difficult direction is the upper bound.

The equality $\beta(G, w) = \alpha(G, w)$ for every nonnegative w is called $\hbar$ -perfection [4]. Known sufficient classes do not settle all line graphs. Line graphs are also the natural frustration graphs of quadratic Majorana Hamiltonians [2, 3], while their stable sets are precisely matchings. This suggests that a quantum uncertainty body and a classical matching polytope might coincide. The missing issue is not the degree constraints but the entire family of blossom, or odd-set, constraints.

Our contribution is a direct mechanism for all of them. If Q is a skew contraction, every odd principal submatrix loses at least one rank. This turns its Frobenius norm into the corresponding odd-set inequality. Together with row-norm degree constraints, Edmonds' theorem identifies the squared entries of Q as a point of the matching polytope. Weighted Cauchy–Schwarz then closes the quantum bound exactly.

The resulting theorem has three useful forms:

1. every finite line graph is $\hbar$ -perfect;
2. every nonnegative weighted instance is solved by maximum-weight matching, without state simulation;
3. a sharp support-sensitive nuclear-norm inequality holds for all real skew matrices.

The last form distinguishes the result from a merely unweighted skew-energy estimate. We also give an independent proof through known simplicial claw-free rank bounds and exhibit an infinite separation from the $\hbar$ -perfect sufficient class. Finally, rather than reporting only successful theorem checks, we compare the complete mechanism against two explicit weaker baselines and ablate skewness itself.

2 Definitions and the variational reduction

For a finite simple root graph $R = (V, E)$, its line graph $L(R)$ has one vertex for every edge of R ; two vertices are adjacent when their root edges share an endpoint. A matching of R is therefore a stable set of $L(R)$, and

$$\alpha(L(R), w) = \nu(R, w), \quad (2)$$

where $\nu(R, w)$ denotes maximum matching weight.

We use the following elementary variational identity. For fixed observables,

$$\max_{\rho} \sum_i w_i \operatorname{tr}(\rho P_i)^2 = \max_{\|a\|_2=1} \left\| \sum_i a_i \sqrt{w_i} P_i \right\|^2. \quad (3)$$

Indeed, Euclidean duality rewrites the squared norm of the expectation vector as a maximization over a . For fixed a , maximization over density matrices gives the squared spectral norm of the Hermitian linear combination. Compactness allows the two maxima to be interchanged. With $c_i = a_i \sqrt{w_i}$, the constraint is $\sum_i c_i^2 / w_i = 1$. Zero-weight coordinates may be deleted.

For a real matrix M , write $\|M\|_{\text{op}}$ for its operator norm, $\|M\|_F$ for its Frobenius norm, and $\|M\|_*$ for its nuclear norm. If $A = -A^T$, then its nonzero singular values occur in equal pairs.

3 A skew-contraction lemma

Lemma 3.1 (Squared skew contractions lie in a matching polytope). *Let $R = (V, E)$ be finite and simple. If $Q = -Q^\top$ and $\|Q\|_{\text{op}} \leq 1$, define $x_e = Q_{uv}^2$ for every $e = uv \in E$. Then $x \in \text{MATCH}(R)$.*

Proof. Nonnegativity is immediate. For each $v \in V$,

$$\sum_{e \in \delta(v)} x_e \leq \sum_{u \in V} Q_{vu}^2 = \|e_v^\top Q\|_2^2 \leq 1.$$

Now let $S \subseteq V$ have odd cardinality. The principal matrix $Q[S]$ is odd-order and skew-symmetric, hence singular and of rank at most $|S| - 1$. Compression cannot increase operator norm, so every singular value of $Q[S]$ is at most one. Consequently

$$2 \sum_{e \in E(S)} x_e \leq \|Q[S]\|_F^2 \leq \text{rank } Q[S] \leq |S| - 1.$$

Thus x satisfies nonnegativity, all degree inequalities, and all odd-set inequalities. Edmonds' matching-polytope theorem [1] states that these inequalities describe $\text{MATCH}(R)$ exactly. $\square$

The lemma is support-monotone: entries of Q outside E are irrelevant and only strengthen the row and Frobenius estimates. This matters because the nuclear-norm dual optimizer need not share the sparsity pattern of A .

Theorem 3.2 (Sharp weighted skew-matrix inequality). *Let $w_e > 0$, and let $A = -A^\top$ be real and supported on $E(R)$. With $A_e = A_{uv}$ in any fixed orientation,*

$$\left(\frac{\|A\|_*}{2} \right)^2 \leq \nu(R, w) \sum_{e \in E} \frac{A_e^2}{w_e}. \quad (4)$$

The constant is attained for every R, w .

Proof. By nuclear/operator-norm duality, there is a real Q with $\|Q\|_{\text{op}} \leq 1$ and $\langle A, Q \rangle_F = \|A\|_*$. Replacing Q by $(Q - Q^\top)/2$ preserves its pairing with skew A and does not increase its operator norm. Hence take Q skew. Skew symmetry gives

$$\frac{\|A\|_*}{2} = \sum_{e \in E} A_e Q_e.$$

By Lemma 3.1, $x_e = Q_e^2$ belongs to $\text{MATCH}(R)$, so $\sum_e w_e Q_e^2 \leq \nu(R, w)$. Weighted Cauchy-Schwarz yields

$$\left(\sum_e A_e Q_e \right)^2 \leq \left(\sum_e \frac{A_e^2}{w_e} \right) \left(\sum_e w_e Q_e^2 \right),$$

which proves (4). For sharpness, choose a maximum-weight matching M and set $A_e = w_e$ on M and zero elsewhere. Then $\|A\|_*/2 = \sum_{e \in M} w_e = \nu(R, w)$ and both sides equal $\nu(R, w)^2$. $\square$

4 Exact Pauli uncertainty on line graphs

Theorem 4.1 (All line graphs are $\hbar$ -perfect). *For every finite simple graph R and every nonnegative edge weight w ,*

$$\boxed{\beta(L(R), w) = \alpha(L(R), w) = \nu(R, w).} \quad (5)$$

Proof. Delete zero-weight edges. Assign a Majorana operator γ_v to each root vertex and represent $e = uv$ by $P_e = i\gamma_u\gamma_v$. After a Jordan–Wigner transform these are Pauli observables, and two anticommute exactly when the root edges share one endpoint [2]. By representation invariance [4], this realization computes $\beta(L(R), w)$.

For real c_e , define $A = -A^\top$ by $A_{uv} = c_e$ in a fixed orientation. Orthogonal skew normal form gives two-by-two blocks $\begin{pmatrix} 0 & s_j \\ -s_j & 0 \end{pmatrix}$. A compatible orthogonal change of Majorana generators turns $\sum_e c_e i\gamma_u\gamma_v$ into a sum of commuting involutions with coefficients s_j . Hence

$$\left\| \sum_e c_e i\gamma_u\gamma_v \right\| = \sum_j s_j = \frac{\|A\|_*}{2}. \quad (6)$$

Combining (3), Theorem 3.2, and the normalized constraint $\sum_e c_e^2/w_e = 1$ gives $\beta(L(R), w) \leq \nu(R, w)$.

Conversely, a maximum-weight matching is a stable set of $L(R)$. Its Pauli observables commute and have a common eigenstate, whose squared expectations on the matching are all one. Therefore $\beta(L(R), w) \geq \nu(R, w)$. Equation (2) completes the proof. $\square$

Corollary 4.2 (Exact uncertainty body). *For any Pauli realization with frustration graph $L(R)$, every squared expectation profile $y_e = \text{tr}(\rho P_e)^2$ belongs to $\text{MATCH}(R)$, and every matching incidence vector is attainable as a dominated profile. Thus all nonnegative support functions of the downward uncertainty body equal those of $\text{MATCH}(R)$.*

Proof. Theorem 4.1 supplies every nonnegative weighted inequality valid for $\text{MATCH}(R)$. Separation for compact downward-closed subsets of the nonnegative orthant gives membership. A common eigenstate attains value one on any fixed matching; uncontrolled coordinates can be discarded by downward closure. $\square$

The theorem is constructive at the level of values: a maximum-weight matching algorithm returns both the exact quantum optimum and a commuting set attaining it. It does not assert an efficient circuit preparing a chosen common eigenstate under every hardware connectivity model.

5 An independent graph-polyhedral proof

Every line graph is claw-free. For a nonisolated root vertex v , the star $\delta(v)$ is a clique in $L(R)$, and for each edge $uv \in \delta(v)$ its neighbors outside the star are edges incident with u , hence form a clique. Thus each connected line graph admits a simplicial clique. Induced subgraphs of line graphs are again line graphs, so the class is hereditary in the form needed below.

The simplicial claw-free sector theorem of Chapman, Elman, and Mann [3] implies the rank inequality

$$\sum_{e \in U} \text{tr}(\rho P_e)^2 \leq \alpha(L(R)[U]) \quad (7)$$

for every induced support U . One obtains it by expressing the Hamiltonian norm through at most α nonnegative single-particle energies and applying Cauchy–Schwarz; this argument is representation-independent.

Apply (7) first to $U = \delta(v)$. Since this is a clique, $\sum_{e \in \delta(v)} y_e \leq 1$. For an odd root-vertex set S , apply it to $U = E(S)$. Then

$$\sum_{e \in E(S)} y_e \leq \alpha(L(R[S])) = \nu(R[S]) \leq \frac{|S| - 1}{2}.$$

Together with $y \geq 0$, these are exactly Edmonds’ inequalities, so $y \in \text{MATCH}(R)$. This reproves Theorem 4.1. The first proof is retained because it is direct and yields the stronger coefficient-sensitive matrix inequality (4).

6 Strictly beyond the h -perfect sufficient class

Xu et al. show that h -perfect graphs are $\bar{h}$ -perfect [4]. The converse route does not explain Theorem 4.1.

Theorem 6.1 (Infinite strict separation). *For every integer $k \geq 2$, $L(K_{2k+1})$ is $\bar{h}$ -perfect but not h -perfect.*

Proof. The first claim is Theorem 4.1. For the second, assign $x_e = 1/(2k)$ to every edge of K_{2k+1} . Maximal cliques of its line graph are root stars and triangles. Their values are respectively 1 and $3/(2k) \leq 1$. An induced odd cycle of length $\ell \geq 5$ has value $\ell/(2k) \leq (\ell - 1)/2$. Hence x satisfies nonnegativity, all clique inequalities, and all induced odd-cycle inequalities used by the h -perfect relaxation.

However,

$$\sum_e x_e = \frac{\binom{2k+1}{2}}{2k} = \frac{2k+1}{2}, \quad \nu(K_{2k+1}) = k.$$

The full-root odd-set inequality is violated by exactly $1/2$, so the stable-set polytope is strictly smaller than that relaxation. Thus the line graph is not h -perfect. $\square$

The witness shows why triangle and odd-hole checks alone miss the mechanism: the new proof supplies blossom inequalities for arbitrary odd vertex sets, not merely chordless cycles.

7 Baseline comparison and mechanism ablations

The proof identifies a specific chain—skew contraction, squared entries, and the complete matching polytope. We tested whether any named link is dispensable. The benchmark contains every nonempty unlabeled root in the NetworkX graph atlas through order seven. Each of the 1245 roots receives one uniform and four seeded positive integer weight vectors, for 6225 weighted instances. This is exhaustive over the small unweighted root types, but the four nonuniform weights are stress cases rather than a probability model for applications.

For each instance we compare the exact maximum-weight matching against three linear relaxations on the same edge coordinates: degree inequalities only; degree, root-triangle clique, and all simple odd-root-cycle inequalities; and degree plus every odd-root-vertex-set inequality. In line-graph coordinates, the second is the clique-and-odd-hole system used by the h -perfect route, while the third is Edmonds’ full matching polytope.

Table 1: Relaxation value divided by exact maximum matching weight over 6225 weighted instances. “Exact” uses relative tolerance 2×10^{-7} .

Baseline	Mean	Median	95th pct.	Maximum	Exact
Degree only	1.0351	1.0000	1.1667	1.5000	66.89%
Cliques + odd cycles	1.0168	1.0000	1.1667	1.2500	84.69%
All odd sets	1.0000	1.0000	1.0000	1.0000	100.00%

Degree-only is strict on 2061 weighted cases from 797 roots; adding the clique-and-odd-cycle constraints still leaves 953 strict cases from 555 roots. The full odd-set system agrees with the

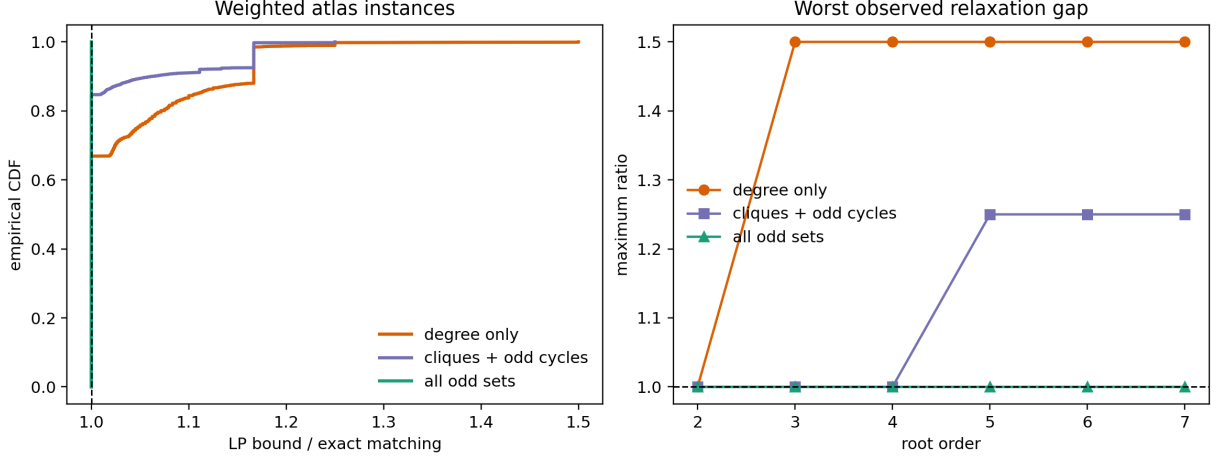


Figure 1: Weaker relaxations systematically overestimate weighted matching. The complete blossom system closes every tested gap. Percentages describe this finite benchmark, not a population of quantum models.

discrete matching value in every instance. The maximum degree gap is the uniform triangle, $3/2$ versus 1. The maximum second-baseline gap is $5/2$ versus 2 on a five-vertex, nine-edge root; the same $5/4$ ratio occurs for K_5 . Figure 1 shows both the aggregate distribution and its dependence on root order.

The bipartite subset is a positive control: all three descriptions are exact on all 710 weighted cases. Among the 5515 nonbipartite cases, degree-only is exact on 62.63%, the clique-and-odd-cycle baseline on 82.72%, and the full polytope on 100%. Thus the implementation recovers the classical bipartite boundary before detecting the missing nonbipartite inequalities.

Controlled families separate the two kinds of odd constraint. With uniform weights on an odd cycle C_n , the odd-cycle constraint repairs the degree-only value $n/2$ to the matching value $(n-1)/2$. On an odd complete root K_n , however, both weaker relaxations retain value $n/2$, exceeding the matching value by $1/2$. We checked $n = 5, 7, 9, 11, 13$. The complete-root blossom, not another local cycle inequality, is the missing object.

Skewness is also necessary. On K_3 , let $Q = I - (2/3)J$. This symmetric Householder matrix has operator norm one and each off-diagonal square is $4/9$. Every degree sum is only $8/9$, but the triangle odd-set sum is $4/3 > 1$. A general contraction can therefore pass all degree tests and fail the matching polytope. In contrast, the skew cross-product matrix generated by $(1, 1, 1)/\sqrt{3}$ has norm one, edge squares $1/3$, and triangle sum exactly one. The odd-dimensional rank loss used in Lemma 3.1 is precisely what disappears in the ablation.

Finally, the sharp constant should not be confused with typical saturation. Across 4980 seeded Gaussian coefficient directions, the median theorem ratio is 0.3992, the 95th percentile is 0.7360, and the maximum is one. After excluding roots with fewer than three edges or matching value below two, the corresponding values are 0.3968, 0.7129, and 0.9997. By contrast, one matching-supported construction for every root gives ratio exactly one in all 1245 cases. The theorem is an exact worst-case support function, not a claim that generic Hamiltonians nearly saturate it. Figure 2 collects these controlled checks.

8 Computational falsification and reproducibility

The proof is arbitrary-size and does not depend on enumeration. We nevertheless implemented a stress audit designed to expose normalization, orientation, and odd-set errors. It evaluates every nonempty graph in the NetworkX graph atlas, giving 1245 root graphs of order at most seven.

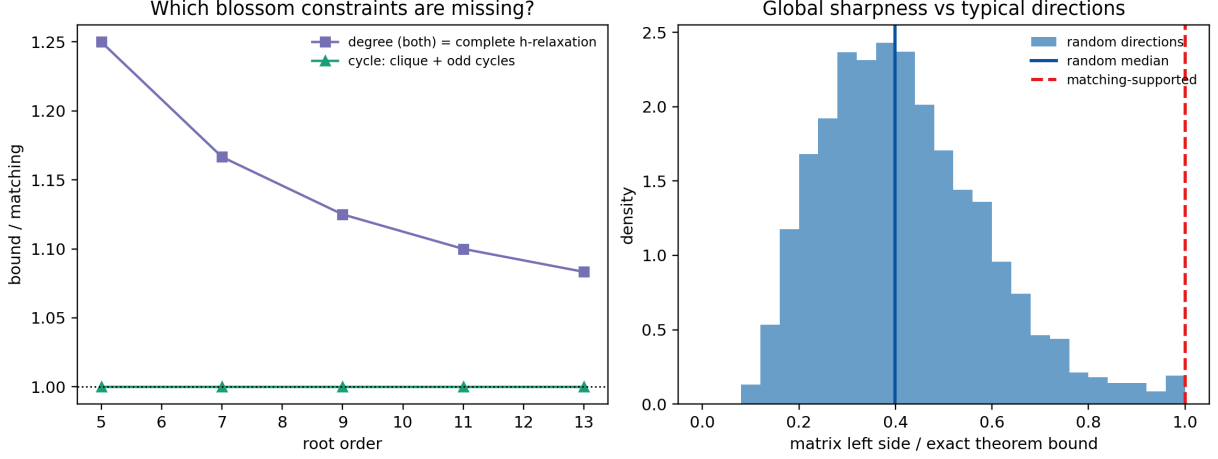


Figure 2: Mechanism ablations. Odd cycles repair cycle roots but not odd complete roots; removing skewness violates the blossom constraint; random directions are usually nonsaturating although equality is globally attainable.

For random positive weights and supported skew matrices, the code independently computes maximum-weight matching, the nuclear-norm expression, a polar dual optimizer, all degree and odd-set residuals, and the theorem ratio.

The largest observed ratio to the proved bound is 1.0000000000000002 . The maximum polar-pairing discrepancy is 3.20×10^{-14} ; maximum degree and odd-set excesses are 1.33×10^{-15} and 2.22×10^{-15} . In 47 small-root cases a dense Jordan–Wigner construction directly checks (6); its maximum error is 7.11×10^{-15} . Exact rational records for K_5, K_7, K_9 check the strict separation. A standard-library verifier checks the frozen summary and source hash, while five tests deliberately corrupt the gap, scope, hash, or norm data and require rejection.

The baseline-and-ablation run is independently frozen in a second JSON file, three CSV tables, and two figures. Its verifier recomputes every aggregate, checks all 6225 full-polytope equalities, proves the Householder counterexample with rational arithmetic, checks the controlled-family formulas, and verifies SHA-256 hashes of all five derived artifacts. Five additional corruption tests require rejection of altered ratios, counterexample entries, hashes, or scope flags.

From the repository root:

```
python experiments/pauli_fourth_moment_phase0/
run_c038_line_graph_hbar.py

python -S experiments/pauli_fourth_moment_phase0/
verify_c038_line_graph_hbar.py

python -m unittest discover -s
experiments/pauli_fourth_moment_phase0
-p test_c038_line_graph_hbar.py -v

python experiments/pauli_fourth_moment_phase0/
run_c039_central_ablations.py -overwrite

python -S experiments/pauli_fourth_moment_phase0/
verify_c039_central_ablations.py

python -m unittest discover -s
experiments/pauli_fourth_moment_phase0
-p test_c039_central_ablations.py -v
```

The wrapped paths are single commands. Numerical agreement is falsification support, not the proof of Theorem 4.1.

9 Relation to prior work and claim boundary

Chapman and Flammia identify line-graph frustration structures as the free fermion setting for quadratic Majorana Hamiltonians [2]. Chapman, Elman, and Mann develop the wider simplicial claw-free framework [3]. Xu et al. define weighted beta numbers and $\hbar$ -perfection and establish other sufficient graph classes [4]; their accessible manuscript does not state the all-line-graph weighted matching equality used here.

The graphs $L(K_{2k+1})$ already appear in magic-state simulation via Jordan–Wigner transformations [5]; the separating graphs themselves are not new. McNulty studies measurement incompatibility on line graphs using skew-energy bounds [6]. Classical skew-energy inequalities related to matching are studied by Tian and Wong [7]. The uniform shadow of (4) is therefore close to known rank and skew-energy bounds. Our priority-sensitive claim is the exact all-weight Pauli beta/matching identity, its complete matching-polytope mechanism, and the sharp weighted matrix form. The underlying ingredients – Edmonds’ theorem, nuclear duality, Majorana normal form, and line-graph solvability – are not claimed new.

The repository contains a source-by-source adversarial audit with explicit overlap and falsification criteria. This supports careful positioning but cannot exclude differently worded or inaccessible prior results. Journal referees remain the independent priority check when no pre-submission expert is available.

10 Conclusion

Odd principal submatrices of a skew contraction lose one rank. That elementary fact generates every blossom inequality and turns the quantum optimization on a line graph into the classical matching polytope. Consequently all finite line graphs are $\hbar$ -perfect for arbitrary nonnegative weights, and the associated skew-matrix inequality is sharp. The infinite family $L(K_{2k+1})$ shows that this mechanism is genuinely broader than the known h -perfect sufficient route.

The ablations sharpen that interpretation. Degree constraints alone explain the entire bipartite control but fail broadly outside it. Adding every clique and odd-cycle inequality reduces rather than eliminates the gap, while full odd-set constraints eliminate it. Removing skewness breaks the key conclusion on the smallest possible odd root. These are finite tests of the proof’s named ingredients, not empirical substitutes for the theorem.

The result concerns exact uncertainty values, not quantum advantage or hardware performance. The next scientific tests are independent refereeing and a wider bibliographic priority search; adding more random instances would not strengthen the analytic theorem.

Data and code availability

Source, frozen records, verifier, and tests are available in the [public C038 GitHub branch](#). The repository index identifies the theorem note, priority audit, generator, frozen JSON, independent verifier, and corruption-controlled tests. Every reported numerical constant is read from the two frozen JSON records by the manuscript checker. The second record includes graph identifiers, all weight vectors and relaxation values, aggregate summaries, explicit scope flags, and hashes for the three CSV tables and two figures.

Declarations

No external funding, competing-interest, authorship, affiliation, or AI-assistance declaration is made in this anonymous working version. These fields must be completed truthfully by the author before submission. No QPU data are used in the theorem or its verification.

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